

Kushagra Jaiswal

ML Engineer & Research Lead

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SUMMARY

ML Engineer building and deploying production ML systems — from research architectures (CNN-Transformer hybrids, CBAM attention) to Kubernetes-served pipelines with MLflow tracking, monitoring, and load-tested serving layers. Focused on shipping work that is verifiable: live deployments, public repositories, and documented evaluation.

EXPERIENCE

ML Engineer & Research Lead

Megaminds IT Services — Remote

May 2025 – Present

- Took research architectures to working implementations: CNN-Transformer hybrids with cross-attention fusion, CBAM spatial/channel attention on ResNet backbones, and federated learning pipelines.
- Reached 97% accuracy on histopathology classification and 89% on multi-class MRI classification.
- Directed technical delivery across a team of 10+ research associates: project assignment, technical mentorship, onboarding, and performance review.
- Handled client communication and requirements directly on delivered projects.
- Domains: medical imaging, NLP classification, RAG pipelines, GAN-based augmentation, AutoML, federated learning.

SELECTED PROJECTS

Retail Demand Forecasting — Kubernetes, MLflow, Production Serving (Personal Project)

Seven-day demand forecasting for 50,000 retail store-product series (FreshRetailNet-50K, CC BY 4.0). LightGBM model with leakage-tested features, enforced by 9 unit tests and a serving-parity check that reproduces all 350,000 evaluation predictions exactly. Scored once on a held-out evaluation week at 0.3317 WAPE: 8.2% better than the strongest classical baseline (0.3612) and 5.1% better than a zero-shot foundation model (Chronos-2, 0.3494).

Served via FastAPI on Kubernetes (kind), with MLflow experiment tracking and model registry, Prometheus monitoring, and locust load testing.

Request batching measured at 3.0x throughput and a 4x lower p99 latency under concurrent load: 82 rps at p50 100 ms on a single CPU pod, verified in-cluster against the running deployment and the service's own Prometheus metrics.

Stack: Python, LightGBM, MLflow, FastAPI, Docker, Kubernetes, Prometheus, locust | github.com/Nerd-coderZero/retail-demand-forecasting

Manga & Comic Translation Pipeline (Personal Project)

End-to-end Chinese-to-English and Japanese-to-English manga translation pipeline: text detection, OCR, LLM-based translation, and in-image text placement.

Hybrid detection/recognition approach for Japanese — PaddleOCR for detection, manga-ocr (purpose-built model) for recognition — after validating that general-purpose OCR performs poorly on vertical manga text.

Three-tier text placement logic that fits translated text at original size, expands into available space when needed, or skips placement rather than force an illegible result.

Deployed on Hugging Face Spaces (Gradio SDK, ZeroGPU); backend built on a FastAPI service layer reused across both the deployed app and an async batch API.

90 automated checks (44 API, 33 browser, 13 interface) run against a stubbed pipeline, with live end-to-end Japanese translation confirmed against the deployed Space.

Stack: Python, PaddleOCR, manga-ocr, NVIDIA NIM (Nemotron), FastAPI, Gradio, Hugging Face Spaces (ZeroGPU) | huggingface.co/spaces/NerdCoderZero/manga-translator

Code: github.com/Nerd-coderZero/Manga-Translator

Agentic Evaluation Harness for LLM-in-the-Loop Pipelines (Personal Project)

LangGraph-based evaluation harness that stress-tests the Manga Translation Pipeline's live deployment: runs batches of real inputs, scores outputs against defined quality checks, and decides to retry, fall back, or flag for review — logging every decision with its reasoning.

Found three real, previously undocumented failure modes on the live system: a garbage-OCR passthrough gap in the upstream filter, an LLM meta-commentary leak, and a content-policy refusal case requiring the pipeline to distinguish a correct model refusal from a translation failure.

All three findings locked in as regression fixtures against real captured data, with a documented design rationale mirroring the source pipeline's own decision-log format.

Stack: *Python, LangGraph, Gradio Client* | github.com/Nerd-coderZero/Manga-Translator-Eval-Harness

eBPF + ML Runtime Threat Detection for Kubernetes (Client Project)

Runtime security system for Kubernetes nodes: an eBPF/BCC node agent streaming syscall and network events to an ML detection service that scores container behaviour for anomalies.

Detection model is an LSTM autoencoder combined with a supervised ensemble, trained and evaluated against a labelled corpus of benign and attack traces spanning several attack classes.

Training pipeline instrumented with MLflow experiment tracking and a model registry; deployed to a Kubernetes cluster with RBAC, configmap and deployment manifests, verified end to end from image build through pod health to live inference on held-out traces.

Stack: *Python, eBPF/BCC, PyTorch, scikit-learn, MLflow, Docker, Kubernetes*

Breast Cancer Detection via Hybrid CNN-Transformer

Histopathology image classification on the BreakHis dataset (7,909 images, 4 magnifications). Fused EfficientNetB3 convolutional features with Transformer cross-attention to capture local texture alongside global context. Beat the CNN-only baseline by 11 percentage points, reaching 97% accuracy on the held-out test set.

Stack: *PyTorch, timm, MONAI*

Brain Tumor Classification with CBAM Attention

4-class MRI classification using ResNet with a Convolutional Block Attention Module (CBAM) for spatial and channel-wise feature weighting. Reached 89% accuracy on the held-out test set.

Stack: *TensorFlow, Keras*

TECHNICAL SKILLS

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, timm, MONAI, LangChain, LangGraph

Architectures: CNN-Transformer hybrids, Vision Transformers (ViT), CBAM attention, Federated Learning, GANs, AutoML

MLOps & Serving: MLflow (tracking & model registry), FastAPI, Docker, Kubernetes, Prometheus, locust

Domains: Medical Imaging, NLP & LLMs, Time-Series Forecasting

Evaluation: ROC/AUC, Macro/Micro F1, WAPE, ablation studies, leakage-parity testing, Optuna tuning, temporal train/test splits

Tools & Infrastructure: Git, REST API, Jupyter, Hugging Face Spaces

EDUCATION

BTech, Computer Science Engineering — J.C. Bose University of Science and Technology, YMCA — 2021

Post Graduate Program, Data Science Bootcamp — UpGrad / Golden Gate University, San Francisco — 2023